

Red Maple

Acer rubrum

Species background: Native to nearly the entire eastern half of North America, from Newfoundland to Florida -- one of the most widely distributed trees on the continent. True to its name, red maple shows red in nearly every season: red flowers in earliest spring, reddish leaf stalks in summer, red fall color, and red twigs in winter.

This packet contains one full lesson for each grade band: K-2, 3-5, 6-8, and 9-12. Each includes a learning objective, standards connection, materials, teacher background, a step-by-step procedure, discussion questions, an assessment, and an extension activity.

GRADE BAND K-2

Red Maple: Find red maple's red clues in whichever season you're observing it.

Standards connection: NGSS K-LS1-1; National Core Arts Standards VA:Cr1 (K-2)

Materials: Coloring sheet, red/orange/yellow crayons.

Teacher background

I might be the most 'red' tree around -- red flowers in spring, red leaf stems in summer, and often the very first bright red leaves of fall.

Procedure

- 1 Engage: Ask students to name things in nature that are red.
- 2 Explore: Look for any red part of the tree available that day -- twig, leaf stalk, or leaf.
- 3 Explain: Explain that this tree shows red in almost every season, which is how it got its name.
- 4 Art: Toss a winged seed and watch it spin (if available), then draw its spinning path with a swirl.
- 5 Evaluate: Student can name at least one red part of the tree and the season it appears.

Discussion questions

Why do you think this tree got the name 'red maple'?

Assessment

Names one correct red feature and its season.

Extension

Look for red maple's spinning seeds on a windy day and watch them fall.

Red Maple: Explain how a spinning samara helps a seed travel farther from its parent tree.

Standards connection: NGSS 3-LS4-3; National Core Arts Standards VA:Cr2 (3-5)

Materials: Coloring sheet, colored pencils.

Teacher background

Find a paired winged seed and toss it -- watch it spin like a helicopter on the way down. Spinning increases a seed's time in the air, giving wind more time to carry it away from the parent tree.

Procedure

- 1 Engage: Ask what would happen to a seed that just dropped straight down versus one that spins.
- 2 Explore: If a samara is available, drop it from shoulder height and watch it spin.
- 3 Explain: Discuss how more time in the air means wind can carry the seed farther.
- 4 Art: Collect fallen leaves in red, orange, and yellow and arrange them into a color gradient from lightest to darkest.
- 5 Evaluate: Student explains in one sentence why spinning helps the seed travel farther.

Discussion questions

Why might traveling farther from the parent tree be good for a new seedling?

Assessment

Correct one-sentence explanation connecting spin time to travel distance.

Extension

Time how long a samara takes to fall from a measured height.

Red Maple: Explain samara aerodynamics and the chemistry behind red fall color.

Standards connection: NGSS MS-LS1-5; National Core Arts Standards VA:Cr2/Re7 (6-8)

Materials: Coloring sheet, leaves in a gradient of colors, notebook.

Teacher background

Spinning increases a seed's time aloft so wind carries it farther. Red fall color comes from anthocyanins the leaf actively makes as chlorophyll breaks down -- unlike yellow, which was there all along, hidden by green.

Procedure

- 1 Engage: Ask why a leaf turning red might be different from a leaf turning yellow, chemically speaking.
- 2 Explore: Collect leaves in a color gradient and sort them from green to fully red.
- 3 Explain: Introduce anthocyanin as a pigment actively made in fall, versus carotenoids (yellow) that were present all along.
- 4 Art: Create a color-wheel-inspired leaf mandala using collected leaves in a gradient, then label which pigment is responsible for each color band.
- 5 Evaluate: Write a short paragraph distinguishing 'pigment that was always there' from 'pigment made fresh in fall.'

Discussion questions

Why would a tree spend energy making a new pigment right before dropping the leaf that holds it?

Assessment

Paragraph correctly distinguishes carotenoids from actively-produced anthocyanins.

Extension

Research why some falls produce more vivid red maple color than others (sunny days, cool nights).

Red Maple: Model samara descent/dispersal and evaluate the energy cost of active anthocyanin production.

Standards connection: NGSS HS-LS1-5; National Core Arts Standards Proficient-level Creating/Responding (9-12)

Materials: Coloring sheet, stopwatch, meter stick, isopropyl alcohol and paper (optional pigment extraction).

Teacher background

Time a seed's fall from a known height to estimate its descent rate, then relate spin-induced autorotation to dispersal distance. Separately, making anthocyanin in fall costs the tree energy even though the leaf is about to drop -- a genuine open question in plant physiology is why that investment might still be worthwhile.

Procedure

- 1 Engage: If making red pigment costs energy right before the leaf drops anyway, what benefit could possibly justify that cost?
- 2 Explore: Time a samara's fall from a measured height and calculate its descent rate; research hypotheses for anthocyanin's fall function (light screening, pest deterrence, nutrient reabsorption protection).
- 3 Explain: Evaluate which hypothesis has the strongest supporting evidence.
- 4 Art: Extract natural pigment from a red maple leaf using isopropyl alcohol and paper chromatography, then mount and label the resulting pigment bands as a scientific illustration plate.
- 5 Evaluate: Submit the descent-rate calculation, a written evaluation of the anthocyanin hypotheses, and the labeled chromatography plate.

Discussion questions

How would you design an experiment to directly test one of the anthocyanin-function hypotheses?

Assessment

Rubric covering calculation accuracy, quality of hypothesis evaluation, and chromatography plate clarity.

Extension

Compare red maple's active fall-color chemistry to sweetgum's, which can show four pigment colors on one tree.

SOURCES & METHODOLOGY

Background research drawn from TreeWalk NYC's own species research. Lesson structure (objective, background, materials, procedure, assessment, extension) follows the format used by real, freely published environmental-education lessons, including the National Park Service's "Looking at Leaves" 5E lesson plan (Acadia Outdoor Classroom Collaborative), Penn State Extension's "Sorting and Classifying Leaves," and Project Learning Tree's leaf-activity guidance -- adapted and written fresh for this species, not copied from those sources. This is an educational pilot; standards references indicate topic alignment, not formal certification.